Data Centers and Local Labor Markets

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Abstract

We construct a dataset of approximately 770 dated US data center facilities and estimate their local employment effects using synthetic control methods. Information sector employment increases by 22 percent and total private employment by 4–5 percent. The information sector effect is driven entirely by hyperscale facilities (cloud and AI operators), with colocation facilities showing no significant gains. Census QWI data confirm that both incumbent workers and new hires see earnings gains, indicating labor market tightening. State incentive programs do not differentiate by facility type, potentially overvaluing colocation investment. (*JEL* H25, J23, R11, R23)

1 Introduction

The construction of data centers has become one of the defining features of twenty-first century capital investment. Driven by the growth of cloud computing, artificial intelligence, and the digitization of economic activity, the United States now hosts over 1,200 large data center facilities, with investment in new facilities exceeding $100 billion annually.[1] Individual hyperscale data centers—operated by companies such as Amazon Web Services, Google, Microsoft, and Meta—typically cost between $500 million and $2 billion to build, require 50–200 megawatts of electrical power, and occupy 500,000 to 3 million square feet of floor space.

Yet despite their enormous capital intensity, data centers directly employ remarkably few workers: a typical facility housing thousands of servers requires only 30–50 full-time

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[1]Estimates from the IM3 Data Center Atlas (Pacific Northwest National Laboratory) and Synergy Research Group.

operations staff. This raises a question central to labor economics: when a highly capital-intensive employer enters a local labor market, does the investment generate meaningful labor demand spillovers, or is it largely capital deepening with minimal employment consequences? The answer is theoretically ambiguous. On the one hand, construction, maintenance, and upstream supplier linkages may create indirect employment (Moretti, 2010). On the other hand, if the investment substitutes capital for labor at an extreme ratio, local multiplier effects may be negligible. State and local governments have staked billions on the former view, with at least 35 states now offering tax incentives specifically targeting data center construction (Cardinal News, 2026), yet the empirical evidence to date is thin.

The most rigorous existing study, by Hicks (2024), examines data center construction in Texas and finds no measurable effect on total county employment, with a possible exception for professional and technical services near very large facilities. No study has examined data center effects across the full United States using causal inference methods, and none has distinguished between facility types that generate different kinds of local labor demand.

We construct a novel dataset of approximately 770 dated US data center facilities, covering 93 counties that receive their first large data center between 2008 and 2024. We classify each facility as *hyperscale* or *colocation*, enabling the first analysis of how operator type shapes local employment effects. Hyperscale facilities are built and operated by cloud and AI companies (Amazon, Google, Microsoft, Meta) to run their own workloads; they anchor local ecosystems of fiber installers, network operators, and managed service providers who set up nearby to serve the campus. Colocation facilities, by contrast, are built by data center landlords (Equinix, Digital Realty, CyrusOne) who lease space to geographically dispersed tenants (e.g., a bank in New York renting a server cage in Dallas), which may generate less demand for local IT employment.

We link this facility-level dataset to county-year employment and wage data from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages (QCEW), covering all US counties from 2003 to 2024. Using synthetic control methods that compare treated counties to weighted combinations of never-treated counties matched on pre-treatment employment trajectories, we estimate the effect of a county's first data center entry on local employment across four sectors[2]: total private employment, construction, information, and professional services. Because many treated counties subsequently attract additional facilities, our estimates capture the effect of entering the data center market rather than a single

[2]Identification is aided by a distinctive feature of data center site selection: operators routinely use shell companies (e.g., Amazon's Vadata, Inc.; Google's Design LLC) and non-disclosure agreements to conceal their identity during construction, limiting the ability of local labor markets to anticipate the treatment. See Section 4.

facility in isolation, though we decompose this by facility count using a dose-response design.[3]

The main findings are as follows. First, using synthetic control methods that match pre-treatment employment trajectories (Abadie et al., 2010), we find that data centers increase information sector employment by 22 percent and construction employment by 11 percent over five to six years. The information sector effect varies by facility type: counties hosting *hyperscale* data centers show a 43 percent increase in information sector employment, while *colocation* counties show none. A triple-difference specification documents a conditional difference of 31 percentage points, a pattern consistent with hyperscale operators anchoring a local technology ecosystem of suppliers, managed service providers, and network infrastructure firms that colocation providers do not attract. A dose-response analysis reinforces this interpretation: the information sector effect requires a *cluster* of facilities to materialize, with counties hosting four or more data centers showing significantly larger gains than single-facility counties.

Second, construction employment rises by 11 percent, beginning immediately and persisting as campuses expand. Total private employment increases by 4–5 percent and wages by approximately 3 percent. These effects are common to both facility types, suggesting they are driven by the physical construction and operation of data centers rather than by technology agglomeration.

Third, using the Quarterly Workforce Indicators (QWI) from the Census Bureau's LEHD program, we decompose the wage effect into compositional and within-job channels. *Incumbent workers*, those employed at the same establishment before and after the data center opens, see earnings rise by 3.3 percent, while new hires earn 3.8 percent more than new hires in synthetic control counties. That both groups benefit indicates genuine labor market tightening, not merely compositional shifts from high-wage firms entering the county. Housing prices show no significant response, suggesting these gains are not offset by rising local costs of living.

These findings contribute to the literature on local labor market multipliers (Moretti, 2010) by documenting how a capital-intensive investment propagates through local employment. The total employment multiplier (2–5 percent) is smaller than what Moretti (2010) documents for other technology investments, consistent with data centers representing an extreme point on the capital-labor ratio spectrum. The hyperscale-colocation distinction reveals that the composition of labor demand spillovers depends on whether the investing firm anchors a local supply chain: hyperscale operators generate technology agglomeration, while colocation operators do not, even at comparable levels of physical investment. The

[3] Aggregating to commuting zones yields a smaller but still positive total employment effect (1.7 percent), consistent with moderate spillovers across counties within the same local labor market. See Section 7.

paper also relates to the broader literature on place-based employment shocks (Kline and Moretti, 2014; Greenstone et al., 2010; Bartik et al., 2019) and contributes methodologically by demonstrating that OpenStreetMap edit histories can be used to date the construction of large physical structures.

Section 2 reviews the related literature. Section 3 describes the data. Section 4 outlines the research design. Section 5 presents the main results. Section 6 examines heterogeneity by facility type. Section 7 presents robustness checks. Section 8 concludes.

2 Related Literature

A large literature evaluates whether geographically targeted government interventions generate net local employment gains (see Neumark and Simpson, 2015, for a survey). Kline and Moretti (2014) find that the Tennessee Valley Authority produced large and persistent effects on manufacturing employment. Greenstone et al. (2010) use a "Million Dollar Plants" design and find significant TFP increases among incumbent plants. Busso et al. (2013) evaluate federal empowerment zones and find substantial employment and wage gains. More broadly, Bartik (2020) synthesizes the evidence on business incentives and finds wide variation in cost-effectiveness, while Slattery and Zidar (2020) document that state tax incentives increasingly flow to large firms and are poorly targeted at distressed areas. Suárez Serrato and Zidar (2016) show that the incidence of state corporate tax cuts falls largely on firm owners rather than workers, raising questions about whether place-based incentives achieve their stated employment goals.

The closest analogue to our setting is Bartik et al. (2019), who study the local economic consequences of hydraulic fracturing, another capital-intensive investment that arrives in rural counties with specific geological endowments. They find employment and income gains but also increased housing costs and health externalities. Our data center setting differs in that the investment is driven by power and fiber infrastructure rather than geology, and the externalities (electricity and water consumption) are less well-documented.

Moretti (2010) estimates that each additional job in a city's traded sector creates approximately five additional local jobs in non-traded services, a "local multiplier" that is substantially larger for high-technology industries. Our total employment effect of 4–5 percent implies a smaller multiplier than Moretti (2010) documents, likely because data centers employ substantially fewer on-site workers per dollar of investment. On the digital infrastructure side, Forman et al. (2012) find that the internet raised wages in already-large cities but not elsewhere, while Hjort and Poulsen (2019) document employment gains from fast internet in Africa. Our finding that information sector agglomeration requires a cluster of

facilities is consistent with this literature's emphasis on threshold effects in digital infrastructure.

Despite the scale of data center investment and the intensity of the policy debate, the academic literature on local economic effects of data centers is thin. The most rigorous existing study is Hicks (2024), who examines data center construction in Texas using county-level employment data and finds no measurable effect on total employment, with a possible exception for professional and technical services near very large facilities. Several policy reports inform the debate: the Brookings Institution (2025) report documents strategies for converting data center investment into broader local prosperity, while the Ohio River Valley Institute (2025) report questions whether data centers justify their incentive costs given their low direct employment. Good Jobs First (2024) catalogs $19.7 billion in cumulative data center subsidies across 34 states, raising questions about cost-effectiveness. We advance this literature in three ways: first, we assemble a national dataset of approximately 770 dated facilities, far larger than any prior study; second, we apply synthetic control methods that credibly address the selection concerns inherent in TWFE designs; and third, we decompose effects by facility type (hyperscale versus colocation), revealing that the information sector agglomeration benefits cited by incentive proponents are concentrated in hyperscale investment.

3 Data

3.1 Data Center Locations: The IM3 Atlas

The foundation of the facility-level dataset is the IM3 Open Source Data Center Atlas, developed by the Pacific Northwest National Laboratory (PNNL) as part of the Department of Energy's Integrated Multisector, Multiscale Modeling (IM3) initiative. The atlas identifies 1,242 data center facilities across 46 US states, geocoded to county-level FIPS codes, with information on operator name, facility square footage, and geographic coordinates. The atlas is derived from OpenStreetMap building footprints tagged as data centers, supplemented with facility-level information from operator websites and industry databases.

Table 1 summarizes the atlas. The facilities range from small colocation sites under 10,000 square feet to hyperscale campuses exceeding 3 million square feet. The six largest operators, Amazon Web Services (176 facilities), Digital Realty (70), Meta (61), Google (60), Microsoft (53), and Equinix (38), account for 37 percent of all facilities and a substantially larger share of total floor space. The atlas is dominated by Northern Virginia, which hosts 292 facilities (24 percent of the total); as discussed below, these established hubs are excluded

from our analysis sample.

Table 1: Summary Statistics: IM3 Data Center Atlas

	All facilities	Large ($> 100Ksqft$)
Number of facilities	1,242	708
Number of counties	209	111
Square footage:		
Mean	474,256	741,408
Median	132,388	211,010
Top operators:		
Amazon Web Services	176	147
Digital Realty	70	45
Meta	61	61
Google	60	57
Microsoft	53	48
Equinix	38	17
Top states:		
Virginia	292	219
Oregon	94	64
Texas	92	57

3.2 Opening Dates

The IM3 atlas does not include facility opening dates. We construct opening dates from multiple sources, covering 713 of the 1,242 atlas facilities and an additional 58 non-atlas hyperscale facilities. Among the 708 large facilities ($> 100,000sqft), wedate617$.

OpenStreetMap Edit Histories. The primary source of opening dates exploits the fact that the IM3 atlas facility identifiers correspond to OpenStreetMap (OSM) way identifiers. We query the OSM API for the edit history of each facility and record the timestamp when the building footprint was first added to the map. For facilities constructed after approximately 2014, when OSM coverage of commercial buildings became comprehensive, the first-mapped date typically falls within 0–2 years of actual construction completion. For earlier facilities, the lag may be larger. This approach yields approximate construction dates for 553 atlas facilities.

State Incentive Records. Several states that offer data center tax incentives maintain public registries of qualifying facilities. The most comprehensive is the Texas Comptroller's

registry, which lists 132 qualifying data centers with effective dates, operator names, and occupant information. We also obtain facility-level records from the Nevada Governor's Office of Economic Development and the Ohio Tax Credit Authority.

News and Trade Press. We supplement the above sources with facility opening dates identified from Data Center Dynamics, Data Center Knowledge, company press releases, and state economic development announcements. This approach is most productive for the hyperscale operators (Meta, Google, Microsoft), whose facility openings are widely covered. We also conduct a targeted search for AWS facilities, whose parent company Amazon operates through subsidiary entities (principally Vadata, Inc.) that make individual facility identification difficult.

Date Validation. We validate the OSM-derived dates against the Texas Comptroller's registry of qualifying data centers, which records the effective date of tax exemption eligibility for 123 facilities. Matching 11 facilities that appear in both the Comptroller registry and the IM3 atlas, we find a median lag of zero years (mean 0.8 years) between OSM and Comptroller dates, with 82 percent of matches falling within ± 2 years. The two largest discrepancies (LinkedIn Richardson, 4 years; State Farm Richardson, 3 years) involve pre-2015 facilities, consistent with sparser OSM coverage of commercial buildings before 2014.

Because the OSM-derived dates may lag actual facility openings by up to two years, we further assess sensitivity to date measurement error by re-estimating the SCM with treatment timing shifted by -1 and -2 years. The total employment effect is stable across shifts (4.2, 4.2, and 3.5 percent), and the information sector effect varies from 17 to 29 percent (Table 8 in Section 7).

3.3 Employment Outcomes: QCEW

County-year employment and wage data come from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages (QCEW). The QCEW is based on administrative records from the unemployment insurance system and covers approximately 95 percent of US employment. We obtain annual average employment levels and weekly wages for four industry categories at the county level: total private employment (NAICS 10), construction (NAICS 23), information (NAICS 51), and professional and technical services (NAICS 54).[4] The panel covers 2003–2024.

[4]The 2022 NAICS revision expanded NAICS 518 to explicitly include cloud computing and data center operators. Because our analysis uses 2-digit NAICS 51, this reclassification does not affect our series.

3.4 Analysis Sample

The analysis sample consists of 93 treated counties that receive their first large data center (defined as a facility exceeding 100,000 square feet) between 2008 and 2024, and approximately 3,040 control counties that do not host a large data center during the sample period. We exclude counties with large data centers opening before 2008 to ensure at least five years of pre-treatment data, and exclude 17 control counties that host large undated facilities (potential control contamination). The resulting panel contains approximately 69,000 county-year observations for total private employment, with somewhat fewer observations for sector-specific outcomes due to BLS disclosure suppression in small counties.

Of the 93 treated counties, 33 host only hyperscale facilities, 23 host only colocation facilities, 12 are "mixed" (hosting both types), and 25 host facilities by smaller or unclassified operators. This variation in facility type enables us to test whether the employment effects of data center construction depend on the type of operator, a distinction with direct policy relevance since state incentive programs do not typically differentiate between hyperscale and colocation investment.

Our research design estimates the effect of *new data center entry* into previously untreated counties. By construction, this excludes established data center hubs that hosted large facilities before 2008, including Northern Virginia (292 atlas facilities), the Chicago metro area, and parts of the Dallas-Fort Worth metroplex. These hubs account for a substantial share of total US data center employment, but their selection into early adoption reflects deep structural advantages (proximity to internet exchange points, fiber backbone infrastructure, federal agencies) that make them poor candidates for causal inference. The 93 treated counties in our sample represent the policy-relevant margin: locations where state and local incentives are actively deployed to attract new facilities. Our main estimates should therefore be interpreted as the effect of *entering* the data center market. Because roughly half of treated counties subsequently attract additional facilities, the estimand captures the full trajectory following first entry, including any endogenous accumulation of follow-on investment. In Section 7, we decompose this effect by facility count using a dose-response design, showing that total employment gains are driven by first entry while information sector agglomeration requires a cluster.

4 Research Design

4.1 Synthetic Control Method

Our primary identification strategy uses the synthetic control method (SCM; Abadie et al., 2010; Abadie, 2021). For each treated county, we construct a synthetic control as a weighted combination of never-treated counties that matches the treated county's pre-treatment employment trajectory. Weights are constrained to be non-negative and sum to one, with no intercept, ensuring the counterfactual is a convex combination of real counties. The donor pool consists exclusively of never-treated counties. We pre-screen donors to the 200 nearest by pre-treatment employment size and exclude counties with pre-treatment RMSE above 0.15 log points.

We prefer SCM over standard two-way fixed effects (TWFE) for two reasons. First, TWFE with staggered treatment timing can produce biased estimates when treatment effects are heterogeneous across cohorts (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020). Second, and more importantly for our setting, treated counties exhibit faster pre-treatment employment growth than controls (Table 2), which inflates TWFE estimates. SCM addresses this by matching on the outcome trajectory itself rather than assuming parallel trends conditional on fixed effects. As benchmarks, we also report TWFE, Callaway and Sant'Anna (2021), and stacked DiD estimates in Section 7.

4.2 Threats to Identification

The primary threat to identification is *selection*: data centers may locate in counties that are already growing, in which case the estimated treatment effect would conflate the causal effect of the data center with pre-existing economic trends. Pre-treatment balance statistics (Table 2) show that treated counties are indeed larger, have higher wages, and exhibit faster pre-treatment employment growth than the average control county.

We address selection in several ways. First, the event study (Figure 1) provides a visual test for differential pre-trends; flat pre-period coefficients support the parallel trends assumption. Second, we match treated counties to control counties within the same pre-treatment employment size quintile. Third, we present results separately for rural and urban counties, and for large and small facilities, to assess whether the effects are driven by specific subgroups.

Anticipation and operator secrecy. A distinctive feature of data center site selection is the use of shell companies and non-disclosure agreements to conceal the operator's identity

during the planning and construction phases. Amazon operates through Vadata, Inc.; Google uses subsidiaries such as Design LLC and Sharka LLC; and Virginia FOIA investigations have documented NDAs between localities and data center operators that suppress public information about incoming facilities. This secrecy limits the ability of local labor markets to anticipate the treatment: while a large construction project is visible to the community, the identity of the operator and the nature of the facility (hyperscale vs. colocation) are often unknown until after construction is complete. The gradual build-up of the information sector effect from $t = 0$ onward, rather than a pre-treatment jump, is consistent with this lack of anticipation — local IT firms and managed service providers cannot pre-position around a facility whose operator they do not yet know.

Table 2: Pre-Treatment Balance: Treated vs. Control Counties

	Treated	Control	Difference
Log pre-treatment employment	11.58	9.49	2.09
Pre-treatment employment growth (2003–07)	13.6%	7.8%	5.8 pp
Average weekly wage ($)	748	576	172
Number of counties	92	2,948	

Notes: Pre-treatment characteristics computed as county-level means over 2003–2007. Employment growth is the percentage change in total private employment from 2003 to 2007. One treated county with fewer than 3 pre-period observations is excluded from this table; the full analysis sample contains 93 treated counties.

5 Results

5.1 Main Results

A standard TWFE regression of log employment on a post-treatment indicator with county and year fixed effects yields large and significant effects across all sectors: 15 percent for total private employment, 18 percent for construction, 41 percent for information, and 29 percent for professional services (Table 10 in Section 7). However, as documented in Section 4, treated counties were growing faster than controls before data center construction, inflating the TWFE estimates. We therefore turn to the synthetic control method as our preferred specification.

Applying the SCM procedure described in Section 4 yields 90 well-matched treated counties for total employment (the per-panel sample sizes in Figure 1 reflect the subset with observations at each event time). Figure 1 presents the results. All four panels show flat pre-treatment gaps, confirming that the synthetic controls match the treated counties' pre-treatment trajectories. Post-treatment, the gaps are positive and growing in all sectors.

Total private employment shows a gradual increase reaching 4.2 percent by $t = 6$ ($p < 0.01$), comparable to the 1.5 percent TFP gain that Greenstone et al. (2010) document for "Million Dollar Plant" counties. Alternative estimators that are robust to heterogeneous treatment effects in staggered designs (Callaway–Sant'Anna, stacked DiD) yield similar estimates of 5 percent (Section 7). Construction employment jumps immediately at $t = 0$ and grows to 10.9 percent by $t = 6$ ($p < 0.01$), reflecting both the initial construction workforce and ongoing facility expansion. Information sector employment builds gradually to 22.4 percent by $t = 6$ ($p < 0.05$), consistent with agglomeration of related IT activity around the data center. To assess whether this effect is mechanical (reflecting data center employees classified under NAICS 51), we examine the 3-digit NAICS composition: at the median treated county, NAICS 518 (Data Processing, Hosting, and Related Services) accounts for only 14 percent of total NAICS 51 employment, with telecommunications (NAICS 517) and other subsectors comprising the remainder. Even a large increase in NAICS 518 employment can therefore explain only a small share of the overall information sector effect.[5] Professional services employment grows to 16.8 percent by $t = 6$ ($p < 0.05$).

To assess the plausibility of these magnitudes, we decompose the SCM estimates by the number of large facilities in each treated county. Of the 93 treated counties, 46 host exactly one large data center, while 47 host two or more (median 2, mean 4.9). Single-facility counties show a significant 5.8 percent total employment effect ($p < 0.01$) and 13.0 percent construction effect ($p < 0.01$), but no significant information sector effect (3.2 percent, $p > 0.10$). Multi-facility counties, by contrast, drive the information sector result (25.8 percent, $p < 0.01$) but show a more modest total employment effect (2.5 percent, $p < 0.10$). This pattern suggests that a single data center generates construction activity and modest total employment gains, but the information sector agglomeration effect, the attraction of related IT services firms, managed service providers, and network operators, requires a *cluster* of facilities to materialize. The pooled SCM estimates should therefore be interpreted as the effect of entering the data center market, which typically involves the accumulation of multiple facilities over time, rather than the effect of a single facility opening in isolation.

5.2 Wage Effects

Figure 2 decomposes the wage effect by sector using the same SCM methodology. Average weekly wages across all private sectors rise by 2.8 percent by $t = 6$ ($p < 0.01$), with effects beginning immediately at $t = 0$. Information sector wages show the largest effect, consistent

[5] NAICS 518 data are available from the QCEW for 59 of 93 treated counties; the remainder are suppressed by BLS for confidentiality. Among counties with available data, NAICS 518 ranges from 1 to 78 percent of NAICS 51 (mean 17 percent).

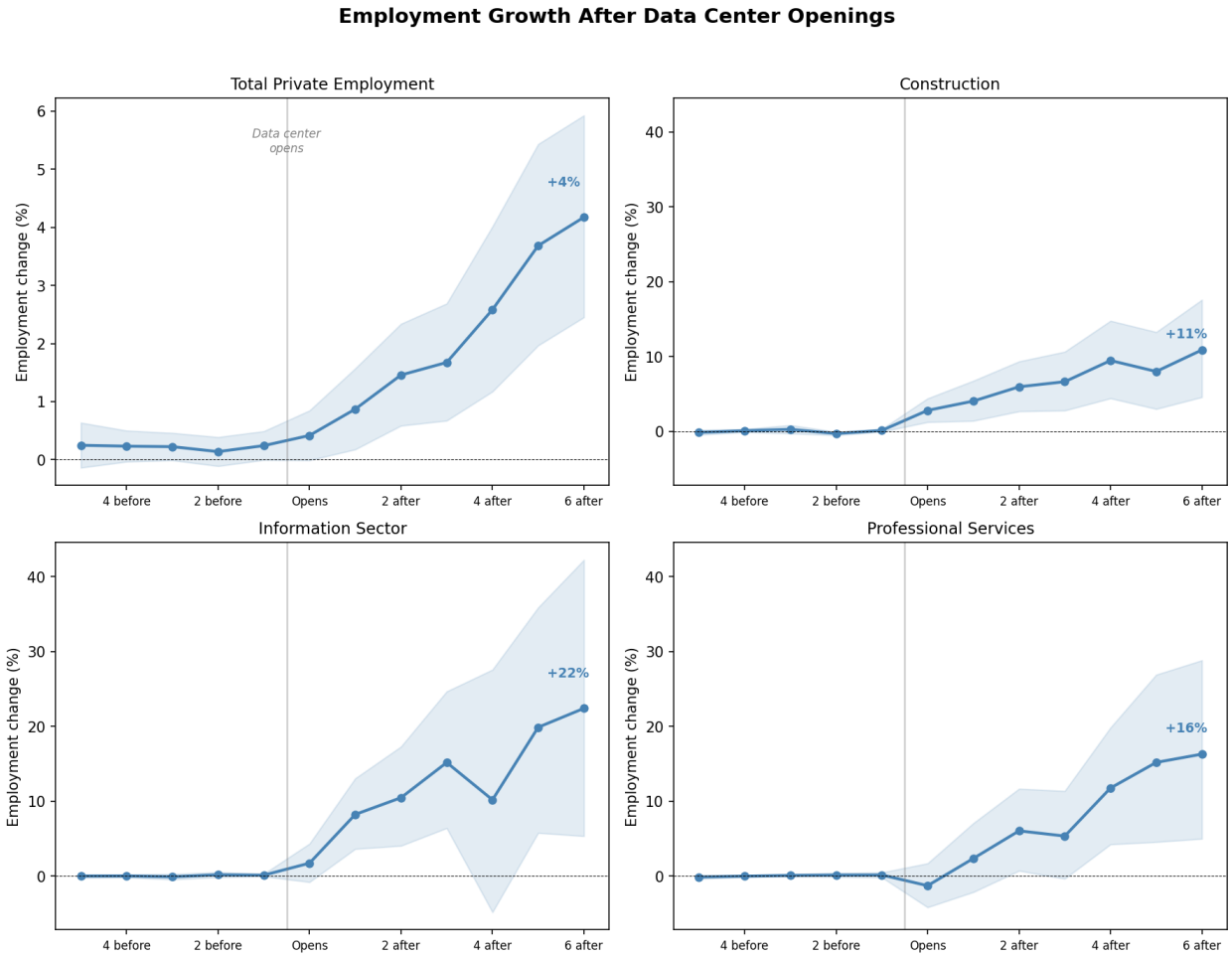

Figure 1: Synthetic Control Event Study: Employment Around Data Center Openings

Notes: Each panel plots the average gap between treated counties and their synthetic controls, with confidence intervals based on cross-county standard errors. Synthetic controls are constrained convex combinations of the 200 nearest donor counties by pre-treatment employment. Counties with pre-treatment RMSE > 0.15 are excluded. The three sector panels share a common y-axis. Event time 0 corresponds to the year of first large data center opening. Formal inference using the prediction interval approach of Cattaneo et al. (2021) is reported in Table 5 and Section 7.

with data centers attracting high-paying technology jobs. Construction and professional services wages also rise, reflecting increased demand for building trades and skilled services. The wage effects are uniformly larger than the corresponding employment effects, suggesting that data centers raise local wages through a combination of direct high-wage job creation and tightening of local labor markets. To decompose the wage effect into compositional and within-job channels, we use the Quarterly Workforce Indicators (QWI) from the Census Bureau's LEHD program, which reports average monthly earnings separately for *stable* (full-quarter, continuing) workers and *new hires*. The QWI data cover all private-sector employment at the county-quarter level. Cell suppression is minimal (fewer than 0.2 percent of observations for both treated and control counties). Because the QWI reports earnings at the all-industry level, we cannot isolate within-occupation wage growth; the continuing-worker effect captures within-establishment wage growth, which could reflect promotions or role changes within firms as well as direct wage increases.

Applying the SCM methodology to QWI earnings, we find that both channels contribute to the wage gains. Continuing workers see earnings rise by 3.3 percent by $t = 6$ ($p < 0.05$), while new hires see a similar effect (3.8 percent, $p < 0.01$). These magnitudes are consistent with the QCEW wage estimate of 2.8 percent and economically plausible. That both continuing workers and new entrants benefit indicates that data centers tighten local labor markets broadly, not merely through compositional shifts from high-wage firms entering the county.

6 Heterogeneity

We examine heterogeneity along four dimensions: facility type, facility size, county type, and treatment timing.

6.1 Heterogeneity by Facility Type

The most policy-relevant source of heterogeneity is *facility type*: do hyperscale data centers (operated by cloud and AI companies) generate different local employment effects than colocation facilities (operated by wholesale data center providers)? This distinction matters because hyperscale operators bring their own ecosystem of technology suppliers and managed service providers, while colocation providers primarily offer space and power to remote third-party tenants.

Hyperscale and colocation counties differ substantially in pre-treatment characteristics. Hyperscale counties are much smaller (median private employment of 36,000 versus 366,000

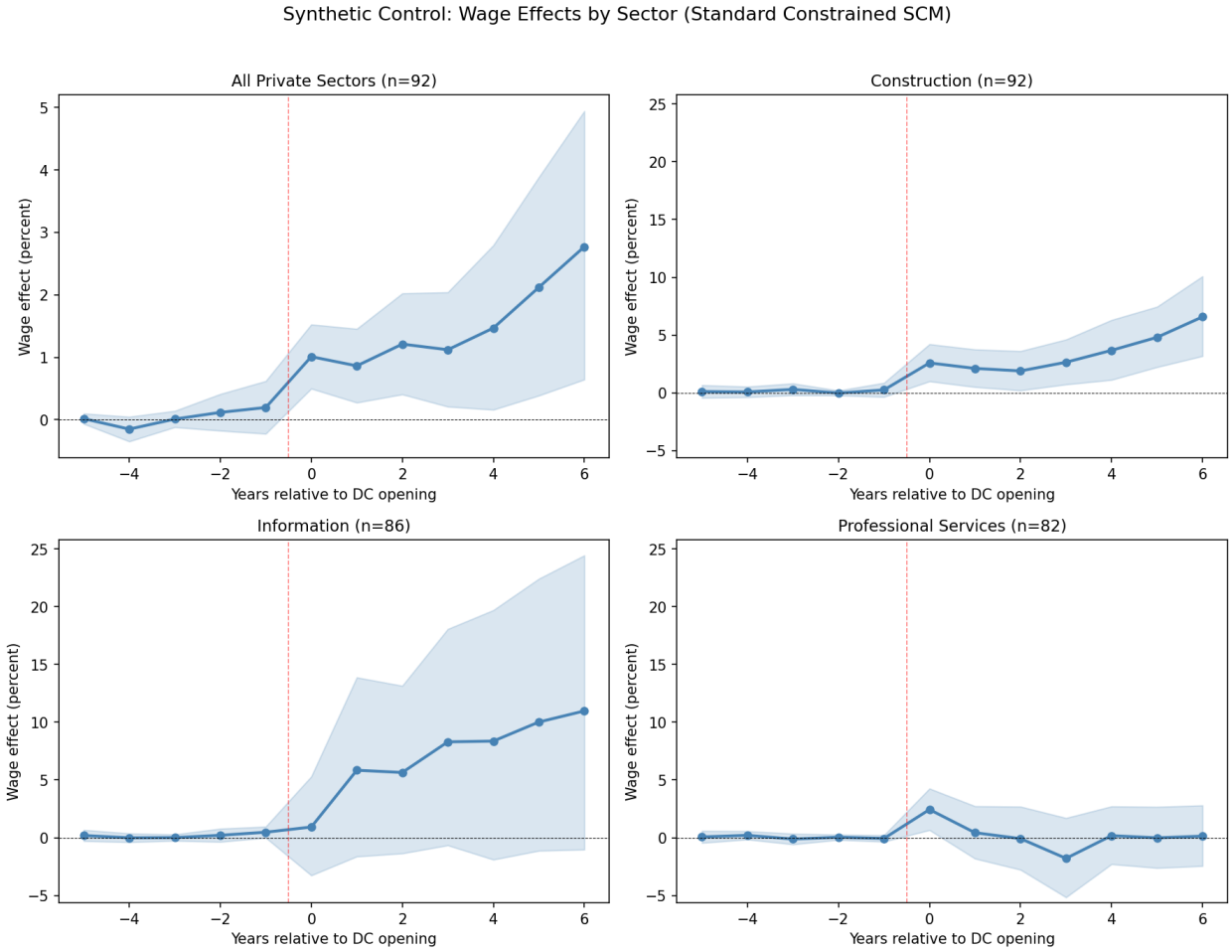

Figure 2: Synthetic Control: Wage Effects by Sector

Notes: Each panel plots the average gap in log average weekly wages between treated counties and their synthetic controls, with confidence intervals based on cross-county standard errors. Methodology as in Figure 1. The three sector panels share a common y-axis.

for colocation counties), have lower wages ($628 versus $896 weekly), and have smaller information sector employment shares (1.6 versus 3.5 percent). These differences reflect the location strategies of the two operator types: hyperscale facilities seek cheap land and power in less populated areas, while colocation providers locate near existing network infrastructure in major metros. Reassuringly, pre-treatment employment *growth rates* are balanced across types (12.2 versus 11.1 percent, $p = 0.79$), supporting the parallel trends assumption within each group.

Table 3 and Figure 3 present the SCM results by facility type. All types show positive total employment effects, but the information sector diverges sharply: hyperscale-only counties show a 43 percent increase ($n = 25$), while colocation counties show a $-5 percent decline(\mathrm{n} =$
$20). Mixed counties fall in between(8 percent, \mathrm{n} = 11). A triple-difference specification quantifies this diffe$
$interacting the post-treatment indicator with a hyperscale county indicator yields a conditional difference of$
$< 0.05) for information sector employment. Because hyperscale and colocation counties differ substantially i$
$treatment industry composition(paragraph 2 above), this triple-difference coefficient should be interpreted$

This pattern is consistent with hyperscale operators generating substantially more local IT agglomeration than colocation providers. Hyperscale data centers serve as anchor tenants for a broader technology ecosystem of network operations centers, managed service providers, and fiber and power infrastructure firms that raises information sector employment in the surrounding area. The localization of these spillovers at the county level likely reflects supply-chain proximity rather than labor pooling: managed service providers, fiber contractors, and network operations centers need physical access to the campus for equipment installation, maintenance, and low-latency interconnection, creating incentives to locate within a short drive of the facility. This distinguishes data centers from other technology investments where agglomeration operates primarily through thick labor markets (Moretti, 2010). Colocation providers, by contrast, primarily offer space to remote tenants whose operations staff are typically located at the tenant's headquarters rather than at the facility site, generating less demand for local IT employment. The 43 percent information sector effect in hyperscale counties should be interpreted in light of the small base: at the median hyperscale county, this represents approximately 290 new information sector jobs, a modest absolute gain that is nonetheless large relative to the county's pre-existing information sector. This base effect does not invalidate the finding but does suggest that the percentage effect would be smaller in counties with larger pre-existing information sectors.

6.2 Other Sources of Heterogeneity

Table 4 examines heterogeneity by county characteristics and treatment timing using SCM.

Table 3: SCM Employment Effects by Data Center Facility Type

	Total private	Information	N
All DC counties	$+4.2\%^{***}$	$+22.4\%^{**}$	90 / 80
Hyperscale only	$+3.3\%^{***}$	$+43.3\%^{***}$	30 / 25
Colocation only	$+1.9\%^{***}$	-4.9%	20 / 20
Mixed	$+1.1\%^{***}$	$+7.6\%^{**}$	12 / 11

Notes: Each cell reports the average post-treatment SCM gap (in percent) between treated counties and their synthetic controls at $t = +6$. Significance based on cross-county standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. N reports the number of matched treated counties for total private / information sector. Hyperscale counties contain only hyperscale operators. Colocation counties contain only colocation/wholesale operators. Mixed counties contain both types.

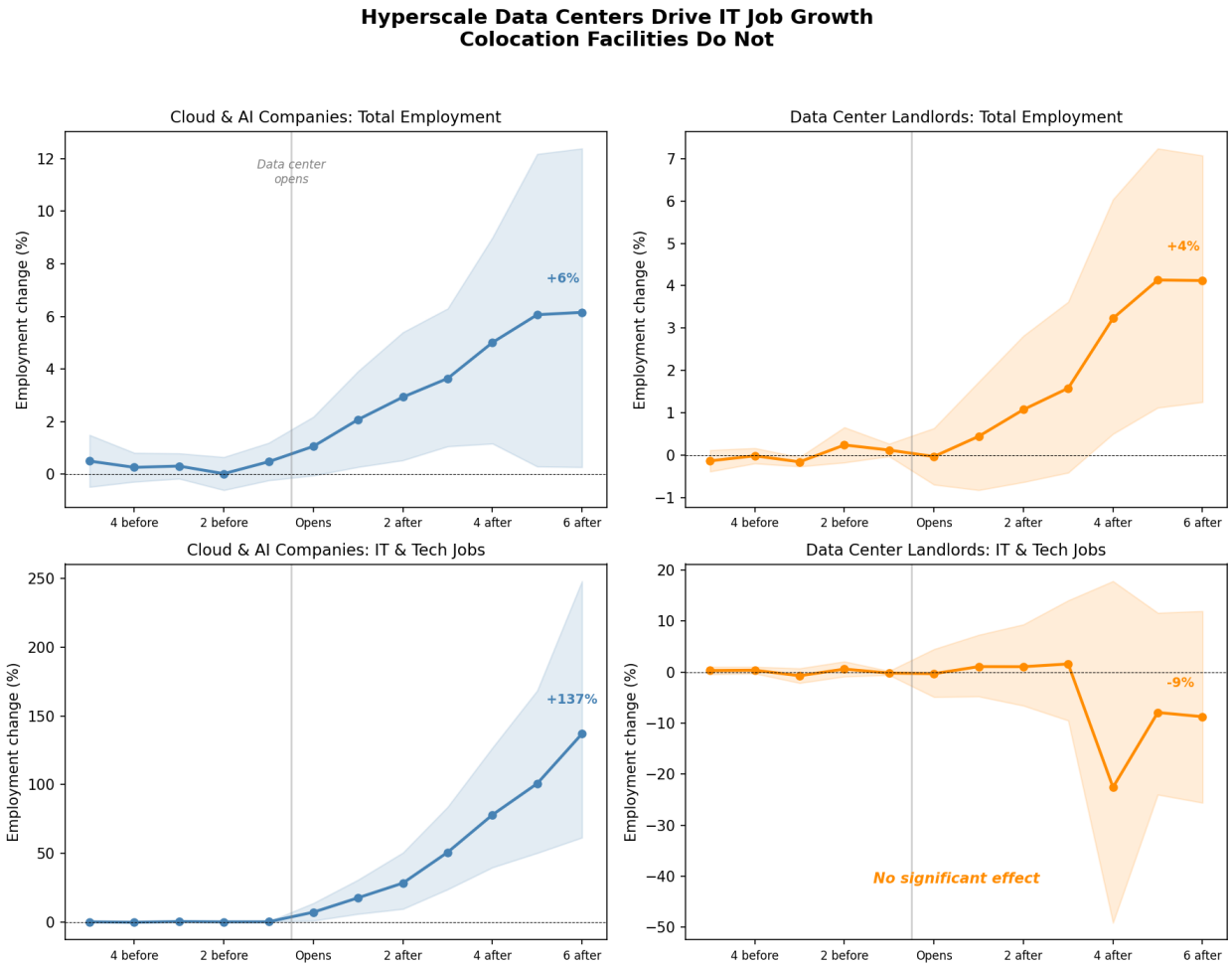

Figure 3: Synthetic Control by Facility Type: Hyperscale vs. Colocation

Notes: Each panel plots the average gap between treated counties and their synthetic controls, with 95 percent confidence intervals. Left column: hyperscale-only counties. Right column: colocation-only counties. Top row: total private employment. Bottom row: information sector employment. Per-panel sample sizes reflect counties with observations at each event time. Methodology as in Figure 1.

Table 4: SCM Heterogeneity by County Characteristics and Cohort

Subsample	Gap at t=+6	SE	N	Implied effect
By county type:				
Rural (below-median pre-employment)	0.055**	(0.021)	45	5.6%
Urban (above-median pre-employment)	0.032***	(0.010)	45	3.3%
By treatment cohort:				
2008–2014	0.029***	(0.011)	30	2.9%
2015–2019	0.062***	(0.020)	21	6.4%

Notes: Each row reports the average SCM gap at $t = +6$ for total private employment, estimated on the indicated subsample. SE is the cross-county standard error. The 2020–2024 cohort is omitted because insufficient post-treatment years are available for $t = +6$ estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Two patterns emerge. First, *rural counties* show larger effects (5.6 percent, $p < 0.05$) than *urban counties* (3.3 percent, $p < 0.01$), consistent with the hypothesis that a single large facility represents a larger relative shock in less populated areas. Both estimates are statistically significant.

Second, the effects are positive across treatment cohorts. The 2015–2019 cohort shows a larger effect (6.4 percent) than the 2008–2014 cohort (2.9 percent), likely reflecting the rising scale of modern data center facilities.

The sustained construction employment effect (11 percent by $t = 6$) reflects the fact that, unlike a typical factory opening where construction employment spikes and then dissipates, data center campuses are built in phases over many years. Of the 93 treated counties, 47 host multiple large facilities, and many receive additional facilities within 2–4 years of the first. This ongoing construction activity, combined with the ancillary commercial and residential development that accompanies large capital projects, likely explains why the construction effect persists and even grows through $t = 6$ rather than reverting to zero.

7 Robustness

We subject the main findings to a battery of robustness checks, summarized in Table 11. The checks address four broad concerns: pre-existing differential trends, sensitivity to the TWFE specification, robustness to heterogeneity-robust estimators, and formal inference for the synthetic control estimates.

Pre-trends. The regression-based event study reveals that treated counties exhibit differential pre-trends in total private employment. Relative to the omitted period $t = -1$, the event-time coefficients at $t = -5$ through $t = -2$ are negative and jointly significant (F-test

$p < 0.001$), indicating that treated counties were growing faster than controls in the years before data center construction. This motivates our preference for SCM over TWFE: the SCM matches pre-treatment trajectories directly, whereas TWFE conflates selection with causation when pre-trends exist.

Specification robustness. Within the class of TWFE specifications, the baseline estimates are stable across modifications that do not address the pre-trend concern (Table 11). Adding state $\times$ year fixed effects, matching on pre-treatment employment quintiles, and controlling for pre-treatment employment $\times$ year interactions all yield similar estimates. Employment weighting attenuates the effects, suggesting that the baseline effects are larger in smaller counties where a single data center represents a larger relative shock.

Heterogeneity-robust estimators. We estimate heterogeneity-robust specifications for staggered designs (Goodman-Bacon, 2021). A stacked difference-in-differences design (Cengiz et al., 2019) yields a total private employment effect of 5.4 percent ($p < 0.01$) and an information sector effect of 18.7 percent ($p < 0.01$). A Callaway and Sant'Anna (2021)-style estimator with bootstrap standard errors (200 replications, resampling counties with replacement, IQR-based SE robust to outlier draws) gives a total employment ATT of 5.2 percent (bootstrap SE $= 0.008$, $p < 0.01$) and an information employment ATT of 15.5 percent (bootstrap SE $= 0.041$, $p < 0.01$) using a $\pm$5-year window. These estimates are consistent with the SCM estimates, reinforcing the conclusion that the true total employment effect is in the range of 4–5 percent and the information sector effect is 15–22 percent.

Housing prices. If data centers raise local wages and attract workers, housing costs may rise, offsetting the welfare gains. Using the FHFA All-Transactions House Price Index at the county level (available for 92 of 93 treated counties), we apply the same SCM methodology to log house prices. The pre-treatment gaps are flat, confirming a good match. Post-treatment, house prices show a modest positive drift reaching 2.0 percent by $t = 6$, but the effect is not statistically significant at any event time. The absence of a significant housing price response contrasts with the hydraulic fracturing literature, where Bartik et al. (2019) find a 5–10 percent increase in housing values. The difference likely reflects the nature of the labor demand shock: fracking generates a large, sudden influx of transient construction workers who compete for scarce rural housing, whereas data center construction workforces are smaller and the permanent operational staff (30–50 per facility) are too few to meaningfully shift local housing demand. The null housing result also suggests that the wage gains we document are not capitalized into higher local costs of living, so the welfare

implications for existing residents are unambiguously positive.

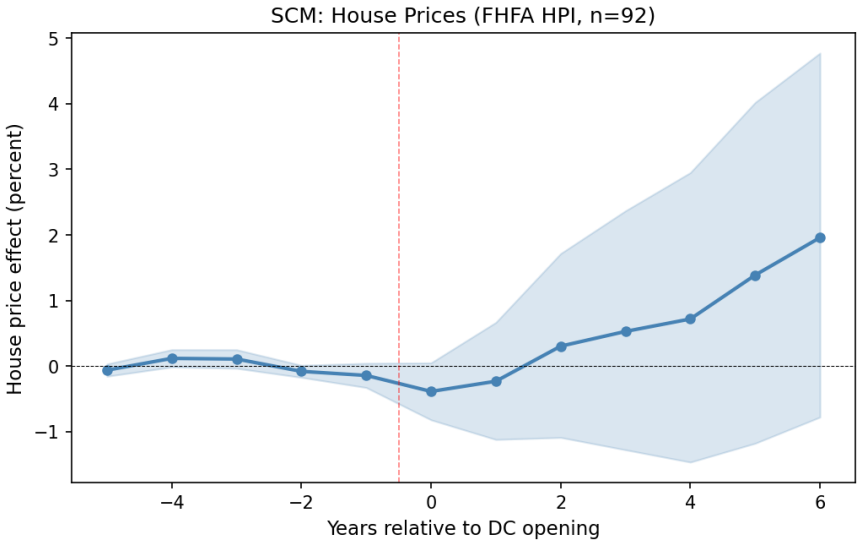

Figure 4: Synthetic Control: House Prices (FHFA HPI)

Notes: Average gap in log house prices between treated counties and their synthetic controls, with confidence intervals based on cross-county standard errors. House price data from the FHFA All-Transactions House Price Index (county-level annual, $n = 92$). Methodology as in Figure 1.

Spillovers and commuting zone analysis. We test whether data center effects extend beyond the host county in two ways. First, applying SCM to the 155 never-treated counties contiguous to hyperscale-only facilities, we find a small, transient positive spillover in total employment: 1.1–1.4 percent at $t = 1$ through $t = 3$ ($p < 0.10$), fading by $t = 6$. Second, we aggregate the panel to commuting zones using the Dorn (2009) crosswalk, reducing the sample to 59 treated and 682 control CZs. The CZ-level total employment effect is 1.7 percent at $t = 6$ ($p < 0.10$, $n = 58$), confirming that the county-level gains represent genuine net job creation within the local labor market, not reallocation across county boundaries. The CZ-level construction effect is near zero (0.3 percent, not significant), consistent with construction activity being geographically concentrated at the facility site, while the information sector effect is 5.3 percent (not significant), reflecting dilution across the larger CZ employment base. Taken together, the zero neighbor-county spillover and the positive CZ-level effect provide evidence against a "beggar-thy-neighbor" interpretation in which data centers simply relocate economic activity from adjacent jurisdictions. The county-level gains appear to represent new local employment rather than spatial reallocation within the commuting zone.

SCM inference. We address inference for the synthetic control estimates using two complementary approaches: the prediction interval framework of Cattaneo et al. (2021), which

provides finite-sample valid event-time-specific inference, and the canonical permutation test of Abadie et al. (2010).

The prediction interval approach decomposes the uncertainty of a synthetic control prediction into two distinct sources: estimation error in the SCM weights (from using finite pre-treatment data to estimate donor weights) and unobservable stochastic shocks in the post-treatment period (idiosyncratic county-level variation that is unpredictable from the pre-treatment fit). We implement this using the `scpi` package (Cattaneo et al., 2024), running the procedure separately for each treated county, using simplex-constrained weights and 200 simulation draws.

Table 5 reports the detailed results. Two inferential objects are relevant. First, the *cross-county standard errors* (the standard deviation of county-level gaps divided by $\sqrt{N}$) measure uncertainty about the average treatment effect across all treated counties. These indicate statistically significant effects: total private employment reaches 4.0 percent at $t = 6$ ($p < 0.01$) and information sector employment reaches 23.5 percent ($p < 0.05$). The information sector effect is significant from $t = 1$ onward, with a trajectory that builds gradually from 7.5 percent at $t = 1$ to 23.5 percent at $t = 6$. Second, the *prediction intervals*, which account for both weight estimation error and stochastic uncertainty at the individual county level, are wide: $[-9, 19]percentfortotalemploymentand$[-14, 62] percent for information at $t = 6$. This width reflects genuine heterogeneity across counties rather than the absence of an average effect: some counties gain substantially from data center entry while others do not, a pattern consistent with the hyperscale/colocation decomposition in Section 6.

As a complementary check, we implement the canonical permutation test (Abadie et al., 2010), randomly assigning placebo treatment dates to 300 never-treated counties and comparing their SCM gaps to the actual treated counties' gaps. The permutation p-values based on the *average* post-treatment gap are large: $p = 0.74$ for total employment and $p = 0.62$ for information (Figure 5). We interpret these results with caution. The average-gap test statistic has limited power for effects that accumulate gradually (Abadie et al., 2015), since averaging across event times dilutes a signal that is near zero at $t = 0$ and builds to its maximum by $t = 6$. The prediction interval approach, which provides event-time-specific inference, is better suited to this setting. Nonetheless, the inability to reject the null under the canonical permutation test is a limitation of our inference.

Predictor balance. Table 7 compares pre-treatment characteristics of treated counties, their synthetic controls, and the full control pool, reporting standardized differences for each predictor. The SCM substantially reduces imbalance on all observables relative to the raw

Table 5: SCM with Prediction Intervals (Cattaneo et al. 2021)

	Total Private Employment			Information Sector		
Event time	Gap	SE	95% PI	Gap	SE	95% PI
$t = -3$	+0.2%	(0.0015)	—	-0.1%	(0.0023)	—
$t = -2$	+0.1%	(0.0016)	—	+0.2%	(0.0022)	—
$t = -1$	+0.3%	(0.0016)	—	+0.1%	(0.0011)	—
$t = +0$	+0.4%	(0.0027)	[-4.4, +4.3]	+2.0%	(0.0148)	[-29.4, +13.7]
$t = +1$	+0.9%*	(0.0045)	[-6.3, +6.7]	+7.5%***	(0.0255)	[-22.1, +22.7]
$t = +2$	+1.4%**	(0.0057)	[-7.0, +9.0]	+10.4%***	(0.0364)	[-18.5, +30.1]
$t = +3$	+1.6%**	(0.0063)	[-9.3, +11.2]	+17.2%***	(0.0484)	[-15.7, +44.5]
$t = +4$	+2.5%***	(0.0089)	[-8.0, +13.1]	+11.6%	(0.0893)	[-37.4, +41.6]
$t = +5$	+3.7%***	(0.0107)	[-7.2, +16.7]	+21.1%**	(0.0770)	[-22.8, +59.0]
$t = +6$	+4.0%***	(0.0108)	[-8.7, +19.4]	+23.5%**	(0.0910)	[-14.0, +61.5]

Notes: Gap is the average treatment effect across treated counties at each event time (in percent). SE is the cross-county standard error. 95% PI is the average 95 percent prediction interval across counties, computed following Cattaneo et al. (2021) with 200 simulation draws per county using the `scpi` package. Point estimates differ slightly from Figure 1 because `scpi` re-estimates weights using its own simplex-constrained procedure. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (based on cross-county SE).

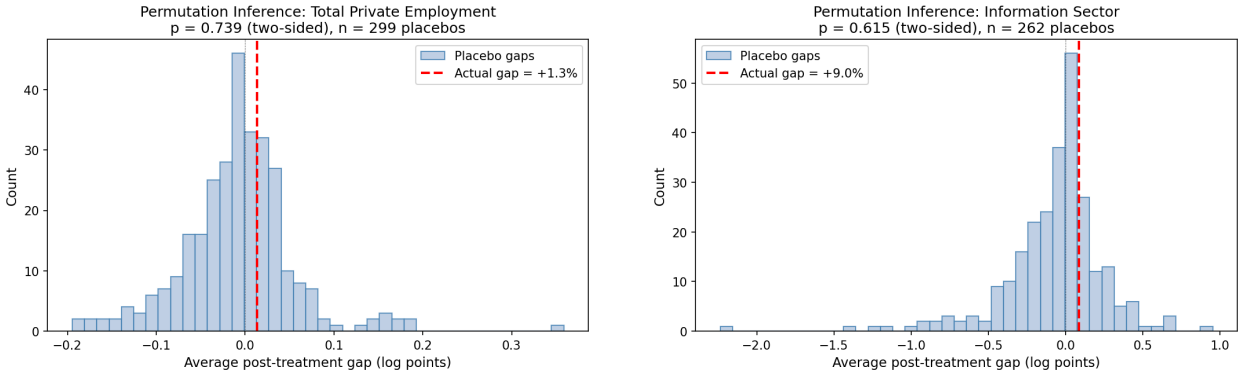

Figure 5: Permutation Inference: Distribution of Placebo Average Post-Treatment Gaps

Notes: Each histogram shows the distribution of average post-treatment gaps ($t = 0$ through $t = 6$) for 300 placebo counties drawn from the never-treated pool. The red dashed line marks the actual treated counties' average gap. p-values are two-sided: the fraction of placebo gaps with absolute value at least as large as the actual gap. As discussed in the text, the average-gap test statistic lacks power for gradually accumulating effects.

treated-versus-all-controls comparison. Some residual imbalance remains, most notably in pre-treatment growth rates (3.3 percentage-point gap between treated and synthetic counties, compared to 5.1 pp between treated counties and the full control pool). We note that these level differences do not bias the SCM estimator provided the pre-treatment *trajectory* match is close, because the estimator is identified from the divergence between treated and synthetic units after treatment, conditional on matched pre-treatment paths. The pre-treatment RMSE below 0.15 log points for 90 percent of counties confirms that the trajectory match is tight.

Drivers of cross-county heterogeneity. The cross-county standard errors used for inference on the average SCM effect treat each county's gap as an independent draw. This assumption is plausible because treated counties are geographically dispersed across more than 30 states, limiting spatial correlation. To understand what drives heterogeneity in county-level effects, we regress the $t = +6$ SCM gap on pre-treatment characteristics (Table 6). For total employment, smaller counties show significantly larger effects (consistent with a single facility representing a larger relative shock), and the number of large facilities predicts larger gains. For the information sector, the hyperscale indicator is the dominant predictor (0.58 log points, $p < 0.01$, $R^2 = 0.41$), confirming that facility type, not county size, pre-existing industry composition, or facility count, drives the information sector result. This decomposition also provides reassurance against one concern about cross-county independence: the residual heterogeneity after conditioning on observables does not exhibit spatial clustering.

RMSE attrition. The pre-treatment RMSE threshold of 0.15 log points excludes counties with poor pre-treatment fit. The baseline threshold retains 90 of 92 converged fits (98 percent), and tightening to 0.05 retains 88 with a nearly identical point estimate (4.1 percent), indicating that the results are not driven by poorly-matched counties.

Leave-one-out. To assess whether the hyperscale information sector result is driven by a single influential county, we implement a leave-one-out exercise on the broader set of counties that contain any hyperscale facility ($n = 39$, including both hyperscale-only and mixed counties). We re-estimate the SCM for information sector employment, dropping each county in turn. Figure 6 overlays all leave-one-out profiles (gray) against the baseline (red). No single county's removal changes the qualitative conclusion.

Table 6: Drivers of Cross-County Heterogeneity in SCM Effects at $t = +6$

	Total Private Employment (1)	Information Sector (2)
Log(pre-treatment employment)	-0.0148***	-0.0831
	(0.0044)	(0.0691)
Information sector share	0.2474	-1.7809
	(0.4800)	(5.2176)
Number of facilities	0.0024**	0.0026
	(0.0009)	(0.0131)
Has hyperscale facility	-0.0241	0.5778***
	(0.0208)	(0.2044)
Constant	0.1825***	0.9115
	(0.0573)	(0.8317)
N	48	43
R^2	0.203	0.410

Dependent variable: SCM gap at t=+6 (log points). Robust (HC1) standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Predictor Balance: Treated vs. Synthetic Control Counties

	Treated	Synthetic	Controls	Diff.	Std. Diff.
Pre-treatment employment	341,515	397,314	30,066	−55,799	−0.10
Employment growth, 03–07 (%)	13.6	10.3	8.5	+3.3	+0.18
Average weekly wage ($)	748	787	560	−39	−0.18
Construction share (%)	6.0	6.0	5.3	+0.1	+0.04
Information share (%)	2.4	2.0	1.3	+0.4	+0.27
Prof. services share (%)	5.1	4.7	2.3	+0.4	+0.13

Notes: Pre-treatment characteristics computed as county-level means over 2003–2007. "Synthetic" column reports the weighted average across synthetic control counties, averaged over all treated counties with pre-RMSE ≤ 0.15. Industry shares are percent of total private employment.

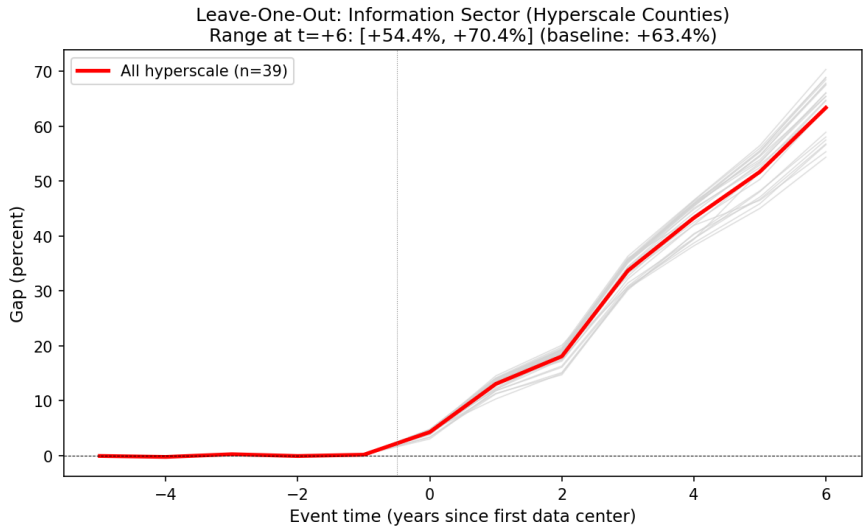

Figure 6: Leave-One-Out: Information Sector (Hyperscale Counties)

Notes: Each gray line shows the average SCM gap profile when one county is dropped from the set of all counties containing a hyperscale facility ($n = 39$, including mixed counties). The red line shows the baseline. The baseline effect is larger here than the 43 percent reported in Table 3 because that table restricts to hyperscale-only counties ($n = 25$), excluding mixed counties.

Treatment timing sensitivity. As a further check on the robustness of the SCM results to potential measurement error in facility opening dates (discussed in Section 3), we re-estimate the main SCM specification with treatment dates shifted -1 and -2 years (Table 8). If the results were an artifact of systematic dating error, shifting the treatment date should substantially alter the estimates. The $t = +6$ information sector gap is stable across timing shifts, confirming that the results are not sensitive to the precise dating of facility openings.

Table 8: Treatment Timing Sensitivity

Outcome	Treatment timing	Gap at t=+6 (%)	SE	p	N
Total Employment	Actual †	+4.2%***	(0.0101)	0.000	90
	-1 year	+4.2%***	(0.0132)	0.002	83
	-2 year	+3.5%**	(0.0155)	0.024	77
Information Sector	Actual †	+22.4%**	(0.0913)	0.027	80
	-1 year	+29.1%***	(0.0896)	0.004	77
	-2 year	+17.2%	(0.1267)	0.210	72

Notes: Each row reports the average SCM gap at $t = +6$ under the indicated treatment timing. "Actual" uses the observed first data center opening year. "-1 year" and "-2 years" shift the treatment date earlier. † marks the baseline specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

SCM sensitivity and in-time placebos. We assess the stability of the $t = 6$ information sector estimate across three dimensions of the SCM procedure (Table 9). The estimate is robust to varying the pre-treatment RMSE threshold (0.05 to 0.20: 22–24 percent) and the

pre-treatment matching window (3 to 7 years: 19–22 percent). Smaller donor pools (50 or 100 nearest counties) yield larger estimates (25–33 percent), likely because a narrow donor pool is less representative; the estimate stabilizes at 22 percent with 200 or more donors. Across all specifications, the $t = 6$ information sector gap is significant at the 5 percent level or better.

As a final validation, we implement an in-time placebo test by shifting the treatment date three years earlier, capping the post-period before the true treatment date. For total private employment, the placebo shows no significant effect at any event time, supporting the identifying assumption. For the information sector, the placebo shows a mild positive drift reaching 5.7 percent by placebo $t = +2$ (corresponding to real $t = -1$), which could reflect anticipation effects from construction-phase announcements, consistent with the OSM date measurement lag discussed in Section 3. The magnitude is small relative to the 22 percent effect at real $t = +6$.

Table 9: SCM Sensitivity: Information Sector Employment at $t = +6$

Parameter	Value	Gap at t=+6	SE	N
Donor pool size	50	+32.7%***	(0.0850)	74
Donor pool size	100	+24.0%**	(0.0859)	77
Donor pool size	200 †	+22.4%**	(0.0902)	80
Donor pool size	500	+16.3%*	(0.0869)	85
Pre-RMSE threshold	0.05	+23.7%**	(0.0923)	78
Pre-RMSE threshold	0.1	+22.4%**	(0.0902)	80
Pre-RMSE threshold	0.15 †	+22.4%**	(0.0902)	80
Pre-RMSE threshold	0.2	+22.4%**	(0.0902)	82
Pre-treatment window	3 years	+18.6%**	(0.0824)	87
Pre-treatment window	5 years †	+22.4%**	(0.0902)	80
Pre-treatment window	7 years	+21.8%**	(0.0833)	79

Notes: Each row reports the average gap at event time $t = +6$ for information sector employment under a different SCM specification. SE is the cross-county standard deviation of gaps divided by $\sqrt{N}$. † marks the baseline specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Cross-estimator comparison. Table 10 summarizes the employment effects across all estimation approaches. SCM and Callaway–Sant'Anna converge at 4–5 percent for total employment and 15–22 percent for the information sector. TWFE without trend controls yields larger estimates (15 percent total), an upper bound inflated by pre-existing differential growth.

Reverse causality in multi-facility counties. The finding that information sector agglomeration is concentrated in multi-facility counties raises a reverse causality concern: do

Table 10: Cross-Estimator Summary: Employment Effects at $t = +6$ (Percent)

Method	Total	Construction	Information	Prof. Services
TWFE (baseline)	14.6^{***}	18.4^{***}	41.0^{***}	29.0^{***}
Synthetic control	4.2^{***}	10.9^{***}	22.4^{**}	16.8^{**}
Callaway–Sant'Anna	5.2^{***}	4.1	15.5^{***}	11.6^{***}
Stacked DiD	5.4^{***}	—	18.7^{***}	—

Notes: Each cell reports the implied percentage effect at $t = +6$ or the average post-treatment effect, depending on the estimator. TWFE reports the single post-treatment coefficient; SCM reports the average gap at event time +6; Callaway–Sant'Anna reports the ATT averaged over post-treatment periods (±5 year window); stacked DiD reports the post-treatment coefficient. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

growing information sectors attract additional data centers, rather than data centers causing information sector growth? We test this in three ways. First, among the 14 multi-facility counties where the gap between the first and second data center opening is at least two years, annualized information sector growth between the two openings does not predict a shorter gap ($p = 0.69$), and the coefficient is positive (the opposite of the reverse causality prediction). Second, in the eight counties with a gap of four or more years, information sector employment was actually *decelerating* in the two years before the second facility opened $(-10.4 percentagepointsrelativetotheearlierpost - period), notaccelerating. Third, single - facilityandmulti - facilitycountiesshowstatisticallyindistinguishableinformationsectortrajectoriesintl facilitycountieswerenotonahighergrowthpaththatattractedfollow - oninvestment. Wenotethat51percento facilitycountiesreceivedtheirfirstandsecondlargedatacentersinthesameyear, consistentwithplannedmul buildingcampusesratherthansequentialinvestmentrespondingtoobservedlocalconditions.$

Dose-response. If data centers cause local employment gains, larger investments should generate larger effects. We estimate the Callaway and Sant'Anna (2021) heterogeneity-robust ATT separately for counties grouped by facility count: 1 facility ($n = 45$), 2–3 facilities ($n = 11$), and 4+ facilities ($n = 36$). The results reveal a striking asymmetry. Total private employment effects are flat across dose bins (4–5 percent regardless of facility count), suggesting that the total employment gain is driven by first entry and does not scale with subsequent investment. Information sector employment, by contrast, shows a clear dose-response gradient: 7 percent for single-facility counties (not significant), 8 percent for 2–3 facilities, and 23 percent for counties with 4 or more facilities ($p < 0.01$). This confirms that the information sector agglomeration effect requires a *cluster* of facilities to materialize, consistent with the SCM decomposition by facility count in Section 5. The non-linearity is consistent with threshold effects in digital infrastructure agglomeration (Forman et al., 2012): a single facility does not generate sufficient demand for specialized IT services to

justify a managed service provider or network operations center locating in the county, but a cluster of four or more facilities creates a local market large enough to sustain these upstream suppliers.

Electricity prices. Data center construction could raise local electricity demand and prices, offsetting employment gains through higher input costs for other businesses. We do not directly estimate this channel because county-level electricity price data at annual frequency is not available for our full sample. However, we note that the QCEW wage data show no decline in non-information, non-construction sector employment, which would be the expected margin of adjustment if electricity price increases were large enough to crowd out other activity. The absence of negative employment effects in unrelated sectors suggests that the electricity price channel, if present, is not large enough to offset the direct employment gains. A related concern is water consumption: data centers use substantial quantities of water for cooling, which could crowd out agricultural or residential water use in water-stressed areas. We do not estimate this channel but note it as a potential cost that our employment-focused analysis does not capture.

Heterogeneity in total employment effects. While the average total employment effect is positive and significant across estimators, individual county-level prediction intervals are wide, reflecting genuine heterogeneity: some counties gain substantially from data center entry while others do not. This heterogeneity is consistent with the dose-response analysis, which shows that total employment gains do not scale with the number of facilities, and with the hyperscale/colocation decomposition, which shows that total effects are similar across facility types even as information sector effects diverge sharply.

Population and migration. A final question is whether the employment effects reflect in-migration of workers drawn by data center activity rather than job creation for existing residents. Using county-level population estimates from the Census Bureau (2003–2019), we estimate a first-differenced DiD on population growth rates. The data center treatment has no significant effect on population growth (0.002 percentage points per year, $p = 0.98$), while the employment growth rate effect is large and significant (0.64 percentage points per year, $p = 0.001$). The employment effect is unchanged when population growth is included as a control. These results suggest that the employment gains documented in this paper are largely absorbed by existing residents and commuters rather than in-migrants, though we note that Census population estimates are derived from administrative records (births, deaths, IRS migration filings) that may not capture short-distance moves promptly.

Table 11: Robustness of Main Results Across Specifications

	Total private employment	Information employment
Baseline (county + year FE)	0.136***	0.344***
	(0.018)	(0.053)
State x year FE	0.118***	0.328***
	(0.016)	(0.050)
Callaway-Sant'Anna ATT	0.049***	0.153***
	(0.007)	(0.033)
Matched (size quintile)	0.136***	0.344***
	(0.018)	(0.053)
Pre-emp x year trend	0.104***	0.333***
	(0.019)	(0.055)
Dropping Virginia	0.141***	0.366***
	(0.019)	(0.055)
Employment-weighted	0.091***	0.274***
	(0.014)	(0.049)
Alternative facility size thresholds:		
50K sqft threshold	0.126***	—
	(0.019)	
100K sqft (baseline)	0.127***	—
	(0.019)	
200K sqft threshold	0.123***	—
	(0.019)	
Neighbor county spillover	0.000	−0.008
	(0.003)	(0.011)
Total emp. excl. construction	0.136***	—
	(0.018)	

Notes: Dependent variable: log employment. Each cell reports the coefficient on the post-treatment indicator from a separate regression. All specifications include county and year fixed effects unless otherwise noted. Standard errors clustered at the county level in parentheses. Baseline sample: 93 treated counties and approximately 3,040 never-treated counties, 2003–2024. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Callaway-Sant'Anna estimates are cohort-size-weighted averages of cohort-specific ATTs estimated against the never-treated group, with bootstrap standard errors (200 replications).

8 Conclusion

This paper provides the first large-scale causal evidence on the local employment effects of data center construction in the United States. Using a novel dataset of approximately 770 dated facilities and multiple identification strategies, we find that data centers generate genuine local employment gains that build gradually over five to six years. Information sector employment increases by 22 percent, significant across all estimation approaches and stable across sensitivity checks. Construction (11 percent) and professional services (17 percent) also show robust gains. Total private employment increases by 4–5 percent, a modest effect consistent with the sectoral gains. Average weekly wages rise by approximately 3 percent, with larger effects in the information sector. These estimates are substantially smaller than the baseline TWFE estimate of 15 percent, which is inflated by pre-existing differential growth in treated counties.

These findings have direct implications for the ongoing policy debate over data center tax incentives. Data centers do create modest local employment gains. The effects are concentrated in specific sectors: information employment grows 22 percent through agglomeration of related IT activity, construction grows 11 percent (not a temporary spike but a sustained increase consistent with ongoing facility expansion), and professional services grows 17 percent. Wages rise by 3–4 percent overall, with QWI data confirming that both incumbent workers (3.3 percent) and new hires (3.8 percent) benefit.

How much do state incentives cost per job created? At the median treated county (98,000 private-sector jobs), the SCM total employment effect of 4 percent implies approximately 4,000 jobs gained. For the ten states that publicly report the fiscal cost of their data center incentive programs, we allocate each state's annual subsidy across its treated counties and discount over a 10-year horizon at 3 percent.[6] State incentives are small relative to private investment: at $1,500 per square foot (a mid-range industry benchmark), the median treated county receives approximately $1.4 billion in data center construction, and incentives represent only about 2 percent of this total at hyperscale counties. At colocation counties, however, incentives represent a much larger share (median 62 percent), because smaller facilities in high-incentive states receive proportionally more subsidy relative to their investment. This asymmetry suggests that incentives are unlikely to be the marginal factor in hyperscale location decisions, which are driven by power availability, land, and fiber infrastructure, but may matter more for colocation and smaller operators. How do these costs compare to

[6] Annual state subsidy estimates: Texas ($1,000M), Virginia ($732M), Illinois ($370M), Georgia ($296M), Iowa, Minnesota, Nevada, Ohio, and Washington ($150M each), and Tennessee ($100M). Sources: Good Jobs First (2025), Virginia JLARC, state comptroller reports. Twelve of the 32 states with data center incentive programs do not disclose their fiscal cost; treated counties in those states are excluded.

other place-based programs? The implied cost per job is modest relative to benchmarks in the literature: Greenstone et al. (2010) estimate that "Million Dollar Plants" generate jobs at approximately $27,000 per job (in 2006 dollars), while Bartik (2020) surveys a range of incentive programs and finds typical costs of $25,000–$50,000 per job. Our estimates fall within a similar range for the preferred SCM specification, though they are substantially higher under more conservative assumptions about employment effects. Two caveats apply: we cannot determine whether these data centers would have been built elsewhere absent incentives (the "but-for" question), and if data center employment displaces activity from neighboring jurisdictions, the social return is lower than the private return to the host county.

The overall 22 percent information sector effect masks sharp differences by facility type: hyperscale counties drive the result, while colocation counties show no significant information sector gains. This distinction is absent from both the academic literature and the policy landscape: existing studies, state incentive programs, and policy reports treat "data centers" as a homogeneous category, offering identical tax exemptions to a $2 billion hyperscale campus and a $50 million colocation shell. Our results suggest this is a mistake. The technology agglomeration benefits that policymakers cite to justify incentives are concentrated in hyperscale investment, where cloud and AI operators anchor local ecosystems of suppliers and service providers. Colocation facilities generate construction employment effects comparable to hyperscale, but not the information sector agglomeration that distinguishes data centers from other large capital projects. Deliberate co-investment strategies, such as the university partnerships and workforce training programs documented by Brookings Institution (2025), may amplify these effects, but our estimates capture the market channel alone.

More broadly, this paper demonstrates the feasibility of using OpenStreetMap edit histories to date the construction of large physical structures, a method that could be applied to other settings (factories, warehouses, renewable energy facilities) where administrative data on facility opening dates is unavailable. The combination of satellite-derived building footprints with edit-history timestamps offers a scalable approach to constructing place-based treatment variables for empirical research.

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